Table 1: Comparison of common nanopore surface modification strategies. Adapted from Eggenberger *et al.*[5] and modified for the design requirements considered in this dissertation.

Method	Phase	Surface bonding	Positive at neutral pH	Diameter control	f
Fluid lipid coatings[6–8]	L/C	Noncovalent	Y	N	
Surfactants[9;10]	L/C	Noncovalent	Y	N	
Polyelectrolyte coatings[11–15]	L	Noncovalent	Y^+	Y	
Self-assembled monolayers[16–21]	L/V	Covalent*	Y^-	N	
ALD	V	Covalent	Y^-	Y^+	
MLD	V	Covalent	N	Y^+	

Phase: L = liquid phase, V = vapor phase, C = colloid.
Scoring: $\mathbf{Y}^+$: excellent or plentiful; $\mathbf{Y}$: at least some suitable options; $\mathbf{Y}^-$: difficult, uncommon, or dubiously suitable; **N**: nominally impossible or unsuitable for hybrid nanopore formation
*Acid-base condensation (as in silanization) is considered a covalent bond, but thiol-on-gold SAMs may be considered noncovalent depending on the environment.